

Linear Algebra 1 2025-26

Unit 8 Worksheet for Week 9 workshops

Linear Algebra of Graphs

Worked solutions to all problems on the worksheet will be made available after the workshop.

Discussion Question 1. Consider the following graph $\mathcal{G}$:

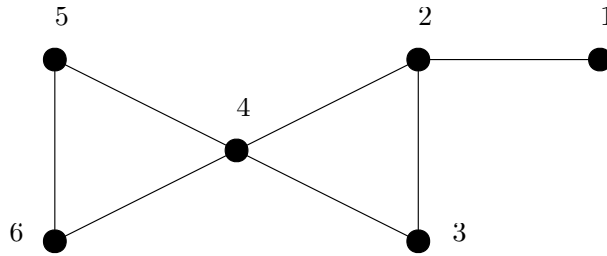


Figure 1: The graph to be used for the discussion tasks.

This exercise is purely a discussion. You need not write down anything for this question.

- (a) What is the shape (i.e. number of rows and columns) of the adjacency matrix $A(\mathcal{G})$? What is the shape of the Laplacian matrix $L(\mathcal{G})$? More generally, what is the shape of those matrices in the case of a generic graph with N nodes?
- (b) What can you say about whether $A(\mathcal{G})$ and $L(\mathcal{G})$ are symmetric, anti-symmetric, or neither? Why? Which property of the graph are you using to determine this?
- (c) What do you obtain if you perform an element-wise sum along the i -th row of $A(\mathcal{G})$? What about the element-wise sum along the j -th column of $A(\mathcal{G})$?
- (d) What do you obtain if you perform an element-wise sum along the i -th row of $L(\mathcal{G})$? What about the element-wise sum along the j -th column of $L(\mathcal{G})$?
- (e) Without doing any computations, guess which node has the highest PageRank score. What about the second highest PageRank score? Explain the reasoning behind your guesses.

Discussion Question 2. Now you will perform the computations to check your intuition from the previous task.

- (a) Write down the adjacency matrix $A(\mathcal{G})$ and the graph Laplacian $L(\mathcal{G})$ of the graph shown in fig. 1.
- (b) Check your answers from discussion Question 1 (a-d) using the matrices you constructed.
- (c) Compute the degree distribution $d(\mathcal{G})$, and sort the nodes from the one with the highest to the lowest degree. Does the degree order agree with your PageRank predictions in discussion Question 1, part (e)?

Discussion Question 3. In this exercise, we'll consider the adjacency matrix $A(\mathcal{G})$ and check it verifies the hypothesis of the Perron-Frobenius theorem.

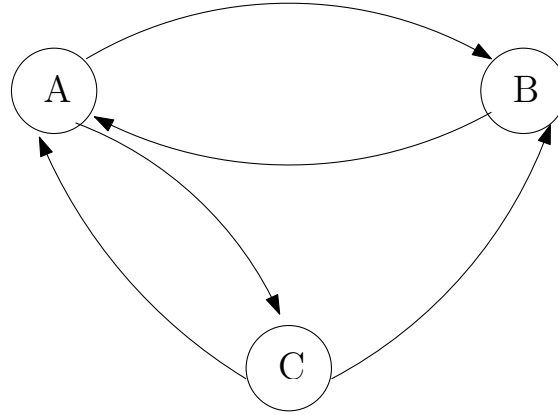
- (a) What is the minimum natural number r (as seen in the Perron-Frobenius Theorem) for which all the entries of A^r are strictly positive? Can you relate this number r with a property of the network?

- (b) Compute the eigenvalues of $A(\mathcal{G})$ ¹. Check that the largest eigenvalue λ_1 is positive with multiplicity 1.
- (c) Check that for all other eigenvalues we have $|\lambda| < \lambda_1$.

Question 1. Compute the PageRank scores of the graph in fig. 1².

Question 2. *The last exercise was good to build some intuition on a reasonably small graph, but the solution may have left you unsatisfied. The graphs was simply too large to write down the PageRank equations and solve them. So let's try something smaller!*

Consider the following directed graph:



- (a) Write down the adjacency matrix $A(\mathcal{G})$ of the graph and the corresponding transition matrix P .
- (b) Write the PageRank equations in the form

$$\mathbf{x} = \gamma P\mathbf{x} + (1 - \gamma) \frac{1}{N} \mathbf{1}\mathbf{x},$$

and express the system as a set of 3 linear equations for x_1, x_2, x_3 .

- (c) Solve the system to find the PageRank vector $\mathbf{x} = [x_1, x_2, x_3]^T$ ³.
- (d) Compute the value for $\gamma = 1$ and $\gamma = 0$ and comment the result.

Question 3. Consider the graph with adjacency matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

- (a) Compute the degree matrix D of the graph.
- (b) Compute the graph Laplacian $L = D - A$.
- (c) Find the Fiedler vector for this graph.

¹You should use a numerical solver for this, do not attempt to find them by pen and paper.

²You may again use a numeric solver for finding the PageRank vector, or ask your tutor for the system solution once you have set up the correct equation

³This is doable pen and paper, but be warned there is a lot of algebra to push through!